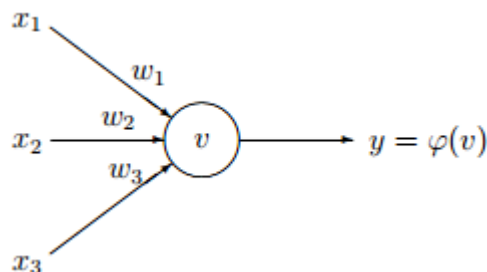


## INT301 Bio-Computation Week 4 Tutorial

**Question 1.** Below is a diagram of a single artificial neuron (unit):



1) It has three inputs  $x = (x_1, x_2, x_3)$  that receive only binary signals (either 0 or 1). How many different input patterns this node can receive? What if the node has four inputs? Five? Can you give a formula that computes the number of binary input patterns for a given number of inputs?

总的输入模式数量是  $2 \times 2 \times 2 = 2^3 = 8$  种

对于给定  $n$  个二进制输入，那么可以接收的不同输入模式总数为  $2^n$ 。

2) Suppose that the weights corresponding to the three inputs have the following values:

$$\begin{aligned} w_1 &= 2 \\ w_2 &= -4 \\ w_3 &= 1 \end{aligned}$$

计算加权和:  $V = w_1x_1 + w_2x_2 + w_3x_3$

将  $V$  应用于阶跃函数  $\phi(V)$  来确定输出  $y$

• 如果  $V \geq 0$ , 则输出  $y=1$

• 如果  $V < 0$ , 则输出  $y=0$

The activation of the unit is given by the step-function:

$$\phi(v) = \begin{cases} 1 & \text{if } v \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Calculate the output value  $y$  of the unit for each of the following input.

$P_1: (x_1=1, x_2=0, x_3=0)$

•  $V = (2 \times 1) + (-4 \times 0) + (1 \times 0) = 2$

•  $V = 2 \geq 0, y=1$

$P_2: (x_1=0, x_2=1, x_3=1)$

•  $V = (2 \times 0) + (-4 \times 1) + (1 \times 1) = 0 - 4 + 1 = -3$

•  $V = -3 < 0, y=0$

| Pattern | $P_1$ | $P_2$ | $P_3$ | $P_4$ |
|---------|-------|-------|-------|-------|
| $x_1$   | 1     | 0     | 1     | 1     |
| $x_2$   | 0     | 1     | 0     | 1     |
| $x_3$   | 0     | 1     | 1     | 1     |

$V \quad -2 \quad -3 \quad 3 \quad -1$

$y \rightarrow \phi(V) \quad 1 \quad 0 \quad 1 \quad 0$

$P_3: (x_1=1, x_2=0, x_3=1)$

•  $V = (2 \times 1) + (-4 \times 0) + (1 \times 1) = 3$

•  $V = 3 \geq 0, y=1$

$P_4: (x_1=1, x_2=1, x_3=1)$

•  $V = (2 \times 1) + (-4 \times 1) + (1 \times 1) = -1$

•  $V = -1 < 0, y=0$

**Question 2.** Assume that you have a single layer perceptron with these weights:  $w = [0.3, -1.2, 0.6]$  初始权重:  $[0.3, -1.2, 0.6]$

$b = 0.1$  (bias) 初始偏置  $b$ : 0.1

Given that you are training the network with the perceptron learning rule with a learning rate of 0.1. What values will the weights become after this positive ( $t = 1$ ) training pattern has been presented to the perceptron (same activation as Q1): 输入向量  $x$ :  $[1.0, 1.0, 1.0]$

$x = [1.0, 1.0, 1.0]$  目标输出  $t$ : 1

Show how you have calculated the new weights. 学习率  $\eta$ : 0.1

$y = \phi(v) = \phi(-0.2) \quad y=0$  阶跃函数  $\phi$ :

权重更新公式:  $w_{new} = w_{old} + \eta(t - y)x$

偏置更新公式:  $b_{new} = b_{old} + \eta(t - y)$

$\phi(v) = 1 \quad \text{if } v \geq 0$

$\phi(v) = 0 \quad \text{if } v < 0$

1. 计算加权和

$V = (w \cdot x) + b$

$V = (w_1x_1 + w_2x_2 + w_3x_3) + b$

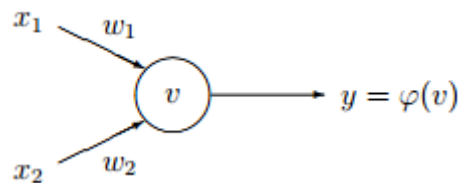
$V = (0.3 \times 1.0) + (-1.2 \times 1.0) + (0.6 \times 1.0) + 0.1$

$V = -0.2$

**Question 3.** Logical operators (i.e. NOT, AND etc) are the building blocks of any computational device. Logical functions return only two possible values, true or false, based on the truth or false values of their arguments. For example, operator AND returns true only when all its arguments are true, otherwise (if any of the arguments is false) it returns false. If we denote truth by 1 and false by 0, then logical function AND can be represented by the following table:

|                          |   |   |   |   |
|--------------------------|---|---|---|---|
| $x_1 :$                  | 0 | 1 | 0 | 1 |
| $x_2 :$                  | 0 | 0 | 1 | 1 |
| $x_1 \text{ AND } x_2 :$ | 0 | 0 | 0 | 1 |

This function can be implemented by a single-unit with two inputs:



with the weights are  $w_1 = 1$  and  $w_2 = 1$  and the activation function is:

$$\varphi(v) = \begin{cases} 1 & \text{if } v \geq 2 \\ 0 & \text{otherwise} \end{cases}$$

Note that the threshold level is 2.

- Test how the neural AND function works.
- Suggest how to change the weights of this single-unit in order to implement the logical OR function (true when at least one of the arguments is true):

输入:  $x_1, x_2$   
 权重:  $w_1=1, w_2=1$   
 激活函数:  $\varphi(v) = \begin{cases} 1, & \text{if } v \geq 2 \\ 0, & \text{otherwise} \end{cases}$

|                         |   |   |   |   |
|-------------------------|---|---|---|---|
| $x_1 :$                 | 0 | 1 | 0 | 1 |
| $x_2 :$                 | 0 | 0 | 1 | 1 |
| $x_1 \text{ OR } x_2 :$ | 0 | 1 | 1 | 1 |

$$V = w_1 x_1 + w_2 x_2$$

① AND

1. 输入:  $x_1=0, x_2=0$

• 加权和:  $V = (1 \times 0) + (1 \times 0) = 0$

• since  $V = 0 < 2$ ,  $y = 0$

2. input:  $x_1=0, x_2=1$

•  $V = (1 \times 0) + (1 \times 1) = 1$

•  $V = 1 < 2$ ,  $y = 0$

3. input:  $x_1=1, x_2=0$

•  $V = (1 \times 1) + (1 \times 0) = 1$

•  $V = 1 < 2$ ,  $y = 0$

4. input:  $x_1=1, x_2=1$

•  $V = (1 \times 1) + (1 \times 1) = 2$

•  $V = 2 = 2$ ,  $y = 1$

the result is same as  $x_1$  and  $x_2$ :  $[0, 0, 0, 1]$

(2) OR

change the weights  $w_1$  and  $w_2$  to  $\begin{cases} w_1=2 \\ w_2=2 \end{cases}$  and keep the activation function

$$w_1=2, w_2=2$$

$$\text{activation function } p(V) = \begin{cases} 1, & \text{if } V \geq 2 \\ 0, & \text{otherwise} \end{cases}$$

1. input:  $x_1=0, x_2=0$

$$\bullet V = (2 \times 0) + (2 \times 0) = 0$$

$$\bullet V = 0 < 2, y = 0$$

2. input:  $x_1=0, x_2=1$

$$\bullet V = (2 \times 0) + (2 \times 1) = 2$$

$$\bullet V = 2 = 2, y = 1$$

3. input:  $x_1=1, x_2=0$

$$\bullet V = (2 \times 1) + (2 \times 0) = 2$$

$$\bullet V = 2 = 2, y = 1$$

4. input:  $x_1=1, x_2=1$

$$\bullet V = (2 \times 1) + (2 \times 1) = 4$$

$$\bullet V = 4 > 2, y = 1.$$

the result is same as  $x_1 \text{ OR } x_2$ :  $[0, 1, 1, 1]$ .